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# Customer Shopping Behavior Analysis

*End-to-End Data Analytics Portfolio Project*

Python | SQL | Power BI

## 1. Project Objective

The purpose of this project was to perform an end-to-end analysis of customer shopping data for a retail business. The raw dataset was cleaned and processed before being analyzed to identify the key factors influencing revenue and sales performance.

Revenue was examined across three main dimensions: gender, subscription status, and age group. The goal was to determine which customer demographic contributes most to the company's revenue and displays the strongest loyalty. These insights were intended to help the business make more targeted and informed decisions around marketing, product offerings, and customer retention.

## 2. Dataset Overview

The dataset used in this project is an open-source retail customer dataset sourced from a public GitHub repository. It contains transactional and demographic information for 3,900 customers across 18 columns, combining both numeric and categorical (text) data types.

Key columns in the dataset include customer demographics (age, gender, location), product details (item purchased, category, size, color), transaction information (purchase amount, payment method, discount applied), and customer engagement metrics (subscription status, review rating, frequency of purchases, previous purchases).

## 3. Tools & Technology Stack

The following tools were used across the different stages of the project:

- Python (pandas) — used for data cleaning and transformation within a Jupyter Notebook environment.
- SQL — used to query the cleaned dataset and answer specific business questions about customer behavior and revenue.
- Power BI — used to build an interactive dashboard that visualizes the key findings from the analysis.

## 4. Project Workflow

The project followed a structured, step-by-step workflow from raw data to final dashboard.

**Data Loading & Initial Exploration**

The raw CSV dataset was loaded into a Jupyter Notebook using the pandas library. The data was initially explored to understand its structure, identify missing values, and assess data types across all columns.

### Data Cleaning & Transformation

Several cleaning and transformation steps were applied to prepare the data for analysis:

- Null values in the Review_Rating column were filled using the median rating of each product category, rather than the overall mean. This approach was chosen because the median is less sensitive to outliers and provides a more representative fill value at the category level.
- Column names were standardized by replacing spaces with underscores, ensuring consistency and compatibility when loading the data into SQL.
- Customers were grouped into four age categories based on the age ranges present in the dataset, allowing for demographic-level analysis.
- The Frequency_of_Purchases column, which contained text values, was mapped to numeric equivalents and stored in a new column to enable quantitative analysis.
- Duplicate or redundant columns were identified and removed to keep the dataset clean and efficient.

### SQL Analysis

The cleaned dataset was loaded into a SQL database. A total of 10 queries were written to analyze customer behavior, covering areas such as revenue by gender, spending patterns by subscription status, age group distribution, discount dependency by product, and repeat purchase behavior.

### Dashboard Creation (Power BI)

The SQL database was connected to Power BI, where an interactive dashboard was built to visualize the key findings. The dashboard includes charts and filters that allow users to explore the data across different dimensions such as category, gender, and subscription status.

## 5. SQL Analysis

> *This section includes selected SQL queries used during the analysis, along with their output. Each query was written to answer a specific business question about customer behavior and revenue.*

- What is the total revenue generated by male vs. female customers?

| | gender | revenue |
|---|---|---|
| 1 | Female | 75191 |
| 2 | Male | 157890 |

- Which are the top 5 products with the highest average review rating?

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| | item_purchased | Average Product Rating |
|---|---|---|
| 1 | Gloves | 3.86142857142857 |
| 2 | Sandals | 3.844375 |
| 3 | Boots | 3.81875 |
| 4 | Hat | 3.8012987012987 |
| 5 | Skirt | 3.78481012658228 |

- Do subscribed customers spend more? Compare average spend and total revenue between subscribers and non-subscribers.

| | subscription_status | total_customers | avg_spend | tot_revenue |
|---|---|---|---|---|
| 1 | Yes | 1053 | 59.4919278252612 | 62645 |
| 2 | No | 2847 | 59.8651211801897 | 170436 |

- Which 5 products have the highest percentage of purchases with discounts applied?

| | item_purchased | discount_rate |
|---|---|---|
| 1 | Hat | 50 |
| 2 | Sneakers | 49 |
| 3 | Coat | 49 |
| 4 | Sweater | 48 |
| 5 | Pants | 47 |

- Segment customers into New, Returning, and Loyal based on their total number of previous purchases, and show the count of each segment.

| | customer_segment | Number of customers |
|---|---|---|
| 1 | Loyal | 3116 |
| 2 | New | 83 |
| 3 | Returning | 701 |

- Are customers who are repeat buyers (more than 5 previous purchases) also likely to subscribe?

| | subscription_status | repeat_buyers |
|---|---|---|
| 1 | No | 2518 |
| 2 | Yes | 958 |

- What is the revenue contribution of each age group?

| | age_group | total_revenue |
|---|---|---|
| 1 | Young Adult | 62143 |
| 2 | Middle-aged | 59197 |
| 3 | Adult | 55978 |
| 4 | Senior | 55763 |

## 6. Power BI Dashboard

> *The dashboard below was built in Power BI to provide an interactive summary of the analysis findings. It covers revenue breakdowns by gender, subscription status, age group, and product category.*


## 7. Key Findings

The analysis produced the following insights about customer shopping behavior and revenue:

- Male customers generate more total revenue than female customers; however, this is largely a reflection of the current customer base composition rather than individual spending differences.
- The average purchase value of subscribed and non-subscribed customers is similar, suggesting the subscription plan does not provide enough financial benefit to significantly influence spending behavior.
- Non-subscribed customers generate more overall revenue and account for a larger portion of repeat buyers, indicating that the subscription program is not attracting or retaining enough customers.
- The customer base is largely made up of repeat buyers, suggesting a strong baseline of customer loyalty.
- Certain products — including hats, sweaters, coats, sneakers, and pants — show a heavy reliance on discounts to drive sales, meaning they rarely sell at full price.
- The young adult age group makes up the largest portion of the customer demographic.

## 8. Business Suggestions

Based on the findings, the following recommendations are proposed to help the business improve revenue and customer engagement:

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- **Women's clothing & marketing —** Invest in women's clothing and targeted marketing.
- **Improve the subscription package —** The fact that subscribed and non-subscribed customers spend similar amounts, combined with non-subscribers being the majority of repeat buyers, suggests the current subscription offering is not valuable enough for customers. The subscription plan should be revised to offer more meaningful benefits — such as better discounts or exclusive perks — to encourage more customers to subscribe and increase retention.
- **Increase supply of discount-reliant products during peak sale periods —** Products such as hats, sweaters, coats, sneakers, and pants depend heavily on discounts to sell. The business should plan for higher stock levels of these items during key sale periods such as Christmas and seasonal clearance events, when discounts are already being offered. This would allow the business to capitalize on increased demand and move more inventory during periods when margins are already reduced.
- **Keep product offerings aligned with current fashion trends —** With young adults making up the largest segment of the customer base, it is important that the product catalog stays current with fashion trends. Regularly updating inventory to reflect what this demographic is interested in will help maintain engagement and reduce the need for discount-driven sales.